

**APPENDIX B3 – WARNING SIGNS AND OBJECT MARKERS
DIMENSION DESCRIPTIONS**

W1-2: Horizontal Alignment – Curve

W1-2R

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the arrow tip.
- E is the distance from the vertical center of the sign to the arrow tip.
- F is the width of the arrow shaft.
- G is the radius of the centerline of the curve in the arrow.
- H is the distance from the horizontal center of the sign to the vertex of the angle of the curve in the arrow.
- J is the distance from the vertical center of the sign to the vertex of the angle of the curve in the arrow.
- K is the distance from the vertical center of the sign to the centerline of the arrow shaft.
- L is the distance from the horizontal center of the sign to the bottom of the arrow symbol.
- M is the width of the arrowhead.
- N is the corner radius.

W1-2L

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the arrow tip.
- E is the distance from the vertical center of the sign to the arrow tip.
- F is the width of the arrow shaft.
- G is the radius of the centerline of the curve in the arrow.
- H is the distance from the horizontal center of the sign to the vertex of the angle of the curve in the arrow.
- J is the distance from the vertical center of the sign to the vertex of the angle of the curve in the arrow.
- K is the distance from the vertical center of the sign to the centerline of the arrow shaft.
- L is the distance from the horizontal center of the sign to the bottom of the arrow symbol.
- M is the width of the arrowhead.
- N is the corner radius.

W1-3: Horizontal Alignment – Reverse Turn

W1-3R

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the tip of the arrowhead.
- E is the distance from the vertical center of the sign to the tip of the arrowhead.
- F is the width of the arrow shaft.
- G is the distance from the vertical center of the sign to the left edge of the upper vertical portion of the arrow shaft.
- H is the radius of the inner edge of the curves in the arrow shaft.
- J is the radius of the outer edge of the curves in the arrow shaft.
- K is the distance from the vertical center of the sign to the right edge of the lower vertical portion of the arrow shaft.
- L is the distance from the horizontal center of the sign to the bottom of the arrow symbol.
- M is the distance from the horizontal center of the sign to the top of the upper horizontal edge of the arrow shaft.
- N is the width of the arrowhead.
- P is the corner radius.

W1-3L

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the tip of the arrowhead.
- E is the distance from the vertical center of the sign to the tip of the arrowhead.
- F is the width of the arrow shaft.
- G is the distance from the vertical center of the sign to the right edge of the upper vertical portion of the arrow shaft.
- H is the radius of the inner edge of the curves in the arrow shaft.
- J is the radius of the outer edge of the curves in the arrow shaft.
- K is the distance from the vertical center of the sign to the left edge of the lower vertical portion of the arrow shaft.
- L is the distance from the horizontal center of the sign to the bottom of the arrow symbol.
- M is the distance from the horizontal center of the sign to the top of the upper horizontal edge of the arrow shaft.
- N is the width of the arrowhead.
- P is the corner radius.

W1-4: Horizontal Alignment – Reverse Curve

W1-4R

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the arrow tip.
- E is the distance from the vertical center of the sign to the arrow tip.
- F is the distance from the vertical center of the sign to the vertex of the angle of the second curve in the arrow shaft.
- G is the width of the arrow shaft.
- H is the distance from the horizontal center of the sign to the vertex of the angle of the second curve in the arrow shaft.
- J is the distance from the horizontal center of the sign to the vertex of the angle of the first curve in the arrow shaft.
- K is the length of the first straight section in the arrow shaft.
- L is the distance from the vertical center of the sign to the vertex of the angle of the first curve in the arrow shaft.
- M is the radius of each curve in the centerline of the arrow shaft.
- N is the width of the arrowhead.
- P is the corner radius.

W1-4L

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the arrow tip.
- E is the distance from the vertical center of the sign to the arrow tip.
- F is the distance from the vertical center of the sign to the vertex of the angle of the second curve in the arrow shaft.
- G is the width of the arrow shaft.
- H is the distance from the horizontal center of the sign to the vertex of the angle of the second curve in the arrow shaft.
- J is the distance from the horizontal center of the sign to the vertex of the angle of the first curve in the arrow shaft.
- K is the length of the first straight section in the arrow shaft.
- L is the distance from the vertical center of the sign to the vertex of the angle of the first curve in the arrow shaft.
- M is the radius of each curve in the centerline of the arrow shaft.
- N is the width of the arrowhead.
- P is the corner radius.

W1-5: Horizontal Alignment – Winding Road

W1-5R

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the vertical center of the sign to the tip of the arrowhead.
- E is the distance from the tip of the arrowhead to the center of the last curve.
- F is the distance from the vertical center of the sign to the vertex of the angle of the fourth curve.
- G is the radius of each curve in the centerline of the arrow shaft.
- H is the distance from the vertex of the angle of the third curve to the vertex of the angle of the fourth curve.
- J is the distance from the horizontal center of the sign to the vertex of the angle of the third curve.
- K is the distance from the vertical center of the sign to vertex of the angle of the third curve.
- L is the length of the diagonal straight section between the second and third curves.
- M is the distance from the horizontal center of the sign to the vertex of the angle of the second curve.
- N is the distance from the vertical center of the sign to the vertex of the angle of the second curve.
- P is the distance from the center radius of the second curve to the vertex of the angle of the first curve.
- Q is the length of the first vertical straight section of the arrow shaft.
- R is the width of the arrow shaft.
- S is the width of the arrowhead.
- T is the corner radius.

W1-5L

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the vertical center of the sign to the tip of the arrowhead.
- E is the distance from the tip of the arrowhead to the center of the last curve.
- F is the distance from the vertical center of the sign to the center of the last curve.
- G is the radius of each curve in the centerline of the arrow shaft.
- H is the distance from the vertex of the angle of the third curve to the top of the fourth curve.
- J is the distance from the horizontal center of the sign to the vertex of the angle of the third curve.
- K is the distance from the vertical center of the sign to vertex of the angle of the third curve.
- L is the length of the diagonal straight section between the second and third curves.

- M is the distance from the horizontal center of the sign to the vertex of the angle of the second curve.
- N is the distance from the vertical center of the sign to the vertex of the angle of the second curve.
- P is the distance from the center radius of the second curve to the vertex of the angle of the first curve.
- Q is the length of the first vertical straight section of the arrow shaft.
- R is the width of the arrow shaft.
- S is the width of the arrowhead.
- T is the corner radius.

W1-10: Combination Horizontal Alignment/Intersection

W1-10R

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the tip of the arrowhead.
- E is the distance from the vertical center of the sign to the tip of the arrowhead.
- F is the width of the arrow shaft.
- G is the radius of the curve in the centerline of the arrow shaft.
- H is the distance from the horizontal center of the sign to the vertex of the angle of the curve in the arrow shaft.
- J is the distance from the vertical center of the sign to the vertex of the angle of the curve in the arrow shaft.
- K is the distance from the vertical center of the sign to the centerline of the first straight section of the arrow shaft.
- L is the distance from the horizontal center of the arrow to the bottom of the arrow shaft.
- M is the width of the arrowhead.
- N is the distance from the absolute center of the sign to the centerline of the 45-degree line at the horizontal center of the sign.
- P is the distance from the horizontal center of the sign to the upper end of the 45-degree line.
- Q is the width of the 45-degree line.
- R is the corner radius.

W1-10L

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the tip of the arrowhead.
- E is the distance from the vertical center of the sign to the tip of the arrowhead.
- F is the width of the arrow shaft.
- G is the radius of the curve in the centerline of the arrow shaft.
- H is the distance from the horizontal center of the sign to the vertex of the angle of the curve in the arrow shaft.
- J is the distance from the vertical center of the sign to the vertex of the angle of the curve in the arrow shaft.
- K is the distance from the vertical center of the sign to the centerline of the first straight section of the arrow shaft.
- L is the distance from the horizontal center of the arrow to the bottom of the arrow shaft.
- M is the width of the arrowhead.
- N is the distance from the center of the sign to the centerline of the 45-degree line at the horizontal center of the sign.

- P is the distance from the horizontal center of the sign to the upper end of the 45-degree line.
- Q is the width of the 45-degree line.
- R is the corner radius.

W1-10a: Combination Horizontal Alignment/Intersection

W1-10aR

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the tip of the arrowhead.
- E is the distance from the vertical center of the sign to the tip of the arrowhead.
- F is the width of the arrow shaft.
- G is the radius of the curve in the centerline of the arrow shaft.
- H is distance from the horizontal center of the sign to the vertex of the angle of the curve in the arrow shaft.
- J is the distance from the vertical center of the sign to the vertex of the angle of the curve in the arrow.
- K is the distance from the horizontal center of the arrow to the centerline of the first straight section of the arrow shaft.
- L is the distance from the vertical center of the sign to the right edge of the horizontal line.
- M is the distance from the centerline of the first straight section of the arrow shaft to the left edge of the horizontal line.
- N is the distance from the horizontal center of the sign to the bottom of the arrow shaft.
- P is the distance from the horizontal center of the sign to the top of the horizontal line.
- Q the width of the horizontal line.
- R is the distance from the bottom of the horizontal line to the bottom of the arrow shaft.
- S is the width of the arrowhead.
- T is the corner radius.

W1-10aL

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the tip of the arrowhead.
- E is the distance from the vertical center of the sign to the tip of the arrowhead.
- F is the width of the arrow shaft.
- G is the radius of the curve in the centerline of the arrow shaft.
- H is distance from the horizontal center of the sign to the vertex of the angle of the curve in the arrow shaft.
- J is the distance from the vertical center of the sign to the vertex of the angle of the curve in the arrow.
- K is the distance from the horizontal center of the arrow to the centerline of the first straight section of the arrow shaft.
- L is the distance from the vertical center of the sign to the left edge of the horizontal line.
- M is the distance from the centerline of the first straight section of the arrow shaft to the right edge of the horizontal line.

- N is the distance from the horizontal center of the sign to the bottom of the arrow shaft.
- P is the distance from the horizontal center of the sign to the top of the horizontal line.
- Q the width of the horizontal line.
- R is the distance from the bottom of the horizontal line to the bottom of the arrow shaft.
- S is the width of the arrowhead.
- T is the corner radius.

W1-10b: Combination Horizontal Alignment/Intersection

W1-10bR

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the tip of the arrowhead.
- E is the distance from the vertical center of the sign to the tip of the arrowhead.
- F is the width of the arrow shaft.
- G is the radius of the curve in the centerline of the arrow shaft.
- H is distance from the horizontal center of the sign to the vertex of the angle of the curve in the arrow.
- J is the distance from the vertical center of the sign to the vertex of the angle of the curve in the arrow.
- K is the distance from the vertical center of the sign to the centerline of the first straight section of the arrow.
- L is the distance from the horizontal center of the sign to the bottom of the arrow.
- M is the distance from the horizontal center of the sign to the top of the vertical line that merges with the arrow.
- N is the width of the line vertical line that merges with the arrow.
- P is the width of the arrowhead.
- Q is the corner radius.

W1-10bL

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the tip of the arrowhead.
- E is the distance from the vertical center of the sign to the tip of the arrowhead.
- F is the width of the arrow shaft.
- G is the radius of the curve in the centerline of the arrow shaft.
- H is distance from the horizontal center of the sign to the vertex of the angle of the curve in the arrow.
- J is the distance from the vertical center of the sign to the vertex of the angle of the curve in the arrow.
- K is the distance from the vertical center of the sign to the centerline of the first straight section of the arrow.
- L is the distance from the horizontal center of the sign to the bottom of the arrow.
- M is the distance from the horizontal center of the sign to the top of the vertical line that merges with the arrow.
- N is the width of the line vertical line that merges with the arrow.
- P is the width of the arrowhead.
- Q is the corner radius.

W1-10c: Combination Horizontal Alignment/Intersection

W1-10cR

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the tip of the arrowhead.
- E is the distance from the vertical center of the sign to the tip of the arrowhead.
- F is the width of the arrow shaft.
- G is the radius of the curve in the centerline of the arrow shaft.
- H is the distance from the horizontal center of the sign to the vertex of the angle of the curve in the arrow shaft.
- J is the distance from the vertical center of the sign to the vertex of the angle of the curve in the arrow shaft.
- K is the distance from the vertical center of the sign to the centerline of the vertical straight section of the arrow shaft.
- L is the distance from the horizontal center of the sign to the bottom of the arrow shaft.
- M is the distance from the vertical center of the sign to the top left corner of the diagonal line that merges with the arrow.
- N is the distance from the horizontal center of the sign to the top left corner of the diagonal line that merges with the arrow.
- P is the width of the diagonal line that merges with the arrow.
- Q is the width of the arrowhead.
- R is the corner radius.

W1-10cL

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the tip of the arrowhead.
- E is the distance from the vertical center of the sign to the tip of the arrowhead.
- F is the width of the arrow shaft.
- G is the radius of the curve in the centerline of the arrow shaft.
- H is the distance from the horizontal center of the sign to the vertex of the angle of the curve in the arrow shaft.
- J is the distance from the vertical center of the sign to the vertex of the angle of the curve in the arrow shaft.
- K is the distance from the vertical center of the sign to the centerline of the first straight section of the arrow shaft.
- L is the distance from the horizontal center of the sign to the bottom of the arrow shaft.
- M is the distance from the vertical center of the sign to the top right corner of the diagonal line that merges with the arrow.
- N is the distance from the horizontal center of the sign to the top right corner of the diagonal line that merges with the arrow.

- P is the width of the diagonal line that merges with the arrow.
- Q is the width of the arrowhead.
- R is the corner radius.

W1-10d: Combination Horizontal Alignment/Intersection

W1-10dR

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the arrow tip.
- E is the distance from the vertical center of the sign to the arrow tip.
- F is the distance from the vertical center of the sign to the vertex of the angle of the second curve in the arrow shaft.
- G is the width of the arrow shaft.
- H is the distance from the horizontal center of the sign to the vertex of the angle of the second curve in the arrow shaft.
- J is the distance from the horizontal center of the sign to the vertex of the angle of the curve in the arrow shaft.
- K is the length of the first straight section in the arrow shaft.
- L is the distance from the vertical center of the sign to the vertex of the angle of the curve in the arrow shaft.
- M is the radius of each curve in the centerline of the arrow shaft.
- N is the width of the arrowhead.
- P is the distance from the horizontal center of the sign to the top end of the diagonal straight line measured at its centerline.
- Q is the distance from the vertical center of the sign to the top end of the diagonal straight line measured at its centerline.
- R is the width of the straight line.
- S is the corner radius.

W1-10dL

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the arrow tip.
- E is the distance from the vertical center of the sign to the arrow tip.
- F is the distance from the vertical center of the sign to the vertex of the angle of the second curve in the arrow shaft.
- G is the width of the arrow shaft.
- H is the distance from the horizontal center of the sign to the vertex of the angle of the second curve in the arrow shaft.
- J is the distance from the horizontal center of the sign to the vertex of the angle of the curve in the arrow shaft.
- K is the length of the first straight section in the arrow shaft.
- L is the distance from the vertical center of the sign to the vertex of the angle of the curve in the arrow shaft.
- M is the radius of each curve in the centerline of the arrow shaft.

- N is the width of the arrowhead.
- P is the distance from the horizontal center of the sign to the top end of the diagonal straight line measured at its centerline.
- Q is the distance from the vertical center of the sign to the top end of the diagonal straight line measured at its centerline.
- R is the width of the straight line.
- S is the corner radius.

W1-10e: Combination Horizontal Alignment/Intersection

W1-10eR

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the tip of the arrowhead.
- E is the distance from the vertical center of the sign to the tip of the arrowhead.
- F is the distance from the vertical center of the sign to the vertex of the angle of the second curve in the arrow shaft.
- G is the width of the arrow shaft.
- H is the distance from the horizontal center of the sign to the vertex of the angle of the second curve in the arrow shaft.
- J is the distance from the horizontal center of the sign to the vertex of the angle of the first curve in the arrow shaft.
- K is the length of the first vertical straight section of the arrow shaft.
- L is the distance from the vertical center of the sign to the vertex of the angle of the first curve in the arrow shaft.
- M is the radius of each curve in the centerline of the arrow shaft.
- N is the width of the arrowhead.
- P is the distance from the horizontal center of the sign to the top end of the diagonal straight line measured at its centerline.
- Q is the distance from the vertical center of the sign to the top end of the diagonal straight line measured at its centerline.
- R is the width of the diagonal straight line.
- S is the distance from the horizontal center of the sign to the bottom end of the diagonal straight line measured at its centerline.
- T is the distance from the vertical center of the sign to the bottom end of the diagonal straight line measured at its centerline.
- U is the corner radius.

W1-10eL

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the tip of the arrowhead.
- E is the distance from the vertical center of the sign to the tip of the arrowhead.
- F is the distance from the vertical center of the sign to the vertex of the angle of the second curve in the arrow shaft.
- G is the width of the arrow shaft.
- H is the distance from the horizontal center of the sign to the vertex of the angle of the second curve in the arrow shaft.
- J is the distance from the horizontal center of the sign to the vertex of the angle of the first curve in the arrow shaft.

- K is the length of the first vertical straight section of the arrow shaft.
- L is the distance from the vertical center of the sign to the vertex of the angle of the first curve in the arrow shaft.
- M is the radius of each curve in the centerline of the arrow shaft.
- N is the width of the arrowhead.
- P is the distance from the horizontal center of the sign to the top end of the diagonal straight line measured at its centerline.
- Q is the distance from the vertical center of the sign to the top end of the diagonal straight line measured at its centerline.
- R is the width of the diagonal straight line.
- S is the distance from the horizontal center of the sign to the bottom end of the diagonal straight line measured at its centerline.
- T is the distance from the vertical center of the sign to the bottom end of the diagonal straight line measured at its centerline.
- U is the corner radius.

W1-11: Hairpin Curve

W1-IIR

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the is the outer radius of the curve in the arrow shaft, measured from the absolute center of the sign.
- E is the distance from the vertical center of the sign to the centerline of the vertical arrow shaft and to the arrowhead tip, the radius of the curve in the centerline of the arrow shaft, and the distance from the horizontal center of the sign to the bottom of the arrow symbol.
- F is the inner radius of the curve in the arrow shaft.
- G is the width of the arrow shaft.
- H is the distance from the horizontal center of the sign to the base of the arrowhead.
- J is the width of the arrowhead.
- K is the corner radius.

W1-IIL

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the is the outer radius of the curve in the arrow shaft, measured from the absolute center of the sign.
- E is the distance from the vertical center of the sign to the centerline of the vertical arrow shaft and to the arrowhead tip, the radius of the curve in the arrow shaft at its centerline, and the distance from the horizontal center of the sign to the bottom of the arrow symbol.
- F is the inner radius of the curve in the arrow shaft.
- G is the width of the arrow shaft.
- H is the distance from the horizontal center of the sign to the base of the arrowhead.
- J is the width of the arrowhead.
- K is the corner radius.

W2-5: Intersection Warning – Y-Intersection

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the vertical center of the sign to the top of the centerline of the left upper branch of the Y-shaped symbol.
- E is the distance from the vertical center of the sign to the top of the centerline of the right upper branch of the Y-shaped symbol.
- F is the distance from the horizontal center of the sign to the top of the centerline of the upper branches of the Y-shaped symbol.
- G is the distance from the horizontal center of the sign to the intersection of the centerlines of the branches of the Y-shaped symbol.
- H is the distance from the intersection of the centerlines of the branches of the Y-shaped symbol to the bottom of the symbol.
- J is the distance from the centerline of each branch to the left and right edges of each branch of the Y-shaped symbol.
- K is the corner radius.

W2-6: Intersection Warning - Circular

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the top of the symbol to the top of the sign.
- E is the distance from the horizontal center of the sign to the top of the symbol.
- F is the distance from the horizontal center of the sign to the bottom of the symbol.
- G is the distance from the bottom of the symbol to the bottom of the sign.
- H is the distance from the left edge of the symbol to the left edge of the sign.
- J is the distance from the vertical center of the sign to the left edge of the symbol.
- K is the distance from the vertical center of the sign to the right edge of the symbol.
- L is the distance from the right edge of the symbol to the right edge of the sign.
- M is the distance from the horizontal center of the sign to the horizontal center of the symbol.
- N is the distance from the vertical center of the sign to the vertical center of the symbol.
- P is the corner radius.

W2-7: Intersection Warning – Offset Side Roads

W2-7R

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the centerline of each line to its parallel edges.
- E is the distance from the horizontal center of the sign to the top of the vertical line.
- F is the distance from the horizontal center of the sign to the bottom of the vertical line.
- G is the distance from the vertical center of the sign to the left end of the upper horizontal line.
- H is the distance from the vertical center of the sign to the right end of the lower horizontal line.
- J is the distance from the horizontal center of the sign to the bottom of the upper horizontal line.
- K is the distance from the horizontal center of the sign to the top of the lower horizontal line.
- L is the corner radius.

W2-7L

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the centerline of each line to its parallel edges.
- E is the distance from the horizontal center of the sign to the top of the vertical line.
- F is the distance from the horizontal center of the sign to the bottom of the vertical line.
- G is the distance from the vertical center of the sign to the right end of the upper horizontal line.
- H is the distance from the vertical center of the sign to the left end of the lower horizontal line.
- J is the distance from the horizontal center of the sign to the bottom of the upper horizontal line.
- K is the distance from the horizontal center of the sign to the top of the lower horizontal line.
- L is the corner radius.

W2-8: Intersection Warning – Side Roads

W2-8R

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the height of the top portion of the vertical line.
- E is the width of each horizontal line.
- F is the distance from the horizontal center of the sign to the bottom of the upper horizontal line.
- G is the distance from the horizontal center of the sign to the top of the lower horizontal line.
- H is the distance from the vertical center of the sign to the left edge of the vertical line.
- J is the distance from the vertical center of the sign to the right edge of the vertical line.
- K is the distance from the vertical center of the sign to the end of each horizontal line.
- L is the corner radius.

W2-8L

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the height of the top portion of the vertical line.
- E is the width of each horizontal line.
- F is the distance from the horizontal center of the sign to the bottom of the upper horizontal line.
- G is the distance from the horizontal center of the sign to the top of the lower horizontal line.
- H is the distance from the vertical center of the sign to the right edge of the vertical line.
- J is the distance from the vertical center of the sign to the left edge of the vertical line.
- K is the distance from the vertical center of the sign to the end of each horizontal line.
- L is the corner radius.

W3-1: Stop Ahead

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the height of the arrow symbol.
- E is the distance from the bottom of the arrow symbol to the top of the octagon.
- F is the distance from the horizontal center of the sign to the top of the octagon.
- G is the distance from the horizontal center of the sign to the bottom of the octagon.
- H is the border width of the octagon.
- J is the distance from the vertical center of the sign to the left edge of the octagon.
- K is the distance from the vertical center of the sign to the right edge of the octagon.
- L is the distance from the vertical center of the sign to the bottom left vertex of the octagon.
- M is the distance from the vertical center of the sign to the bottom right vertex of the octagon.
- N is the sign corner radius.

W3-2: Yield Ahead

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the height of the arrow symbol.
- E is the distance from the bottom of the arrow symbol to the top of the triangle.
- F is the distance from the vertical center of the sign to the top left and right corners of the red triangle ignoring the corner radius.
- G is the red triangle border width.
- H is the distance from the horizontal center of the sign to the top of the red triangle.
- J is the distance from the horizontal center of the sign to the top of the white triangle.
- K is the corner radius of the white triangle.
- L is the corner radius of the red triangle.
- M is the length of each edge of the red triangle ignoring the corner radius.
- N is the sign corner radius.

W3-5: Reduce Speed Limit Ahead

- A is the length of each side of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the height of the arrowhead.
- E is the distance from the barbs of the arrowhead to the bottom of the arrow symbol.
- F is the distance from the bottom of the arrow symbol to the top of the white panel.
- G is the distance from the top of the first line on the white panel to the top of the panel and from the bottom of the last line on the white panel to the bottom of the panel.
- H is the letter height and FHWA standard font for the first and second lines.
- J is the space between the first, second, and third lines. The bottom of the second line is aligned with the horizontal center of the sign.
- K is the letter height and FHWA standard font for the third line.
- L is the inset from the edge of the white panel to the panel's border.
- M is the border width of the white panel.
- N is the corner radius of the white panel.
- P is the distance from the vertical center of the sign to the left edge of the word on the first line.
- Q is the distance from the vertical center of the sign to the right edge of the word on the first line.
- R is the distance from the vertical center of the sign to the left edge of the word on the second line.
- S is the distance from the vertical center of the sign to the right edge of the word on the second line.
- T is the distance from the vertical center of the sign to the left and right edges of word on the third line.
- U is the distance from the vertical center of the sign to the left and right edges of the white panel.
- V is the sign corner radius.

W4-1: Merge

W4-1R

- A is the length of each side of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the tip of the arrowhead.
- E is the distance from the horizontal center of the sign to the base of the arrow shaft.
- F is the width of the arrow shaft.
- G is the shaft width of the curved branch.
- H is the radius of the first section of the branch.
- J is the radius of the second section of the branch.
- K is the distance from the vertical center of the sign to the vertex of the curve angle of the second section of the branch.
- L is the distance from the vertical center of the sign to the vertex of the curve angle of the first section of the branch.
- M is the distance from the horizontal center of the sign to the vertex of the curve angle of the second section of the branch.
- N is the distance from the horizontal center of the sign to the vertex of the curve angle of the second section of the branch.
- P is the width of the base of the arrowhead.
- Q is the corner radius.

W4-1L

- A is the length of each side of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the tip of the arrowhead.
- E is the distance from the horizontal center of the sign to the base of the arrow shaft.
- F is the width of the arrow shaft.
- G is the shaft width of the curved branch.
- H is the radius of the first section of the branch.
- J is the radius of the second section of the branch.
- K is the distance from the vertical center of the sign to the vertex of the curve angle of the second section of the branch.
- L is the distance from the vertical center of the sign to the vertex of the curve angle of the first section of the branch.
- M is the distance from the horizontal center of the sign to the vertex of the curve angle of the second section of the branch.
- N is the distance from the horizontal center of the sign to the vertex of the curve angle of the second section of the branch.
- P is the width of the base of the arrowhead.
- Q is the corner radius.

W4-3: Added Lane

W4-3R

- A is the length of each side of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the tip of each arrowhead and to the base of the left arrow.
- E is the vertical distance from the horizontal center of the sign to the center of the base of the right arrow.
- F is the vertical distance from the center of the base of the right arrow to the center of the base of the parallel dividing line.
- G is the vertical distance from the base of the left arrow to the base of the left dividing line.
- H is the distance from the vertical center of the sign to the tip of each arrowhead.
- J is the length of each dashed segment of the dividing line and the distance from the horizontal center of the sign to the end of the first segment.
- K is the space between the segments of the dividing line.
- L is the offset distance from the centerline of the curved shaft to the centerline of the right branch of the dividing line.
- M is the radius of the curve in the right arrow shaft
- N is the distance from the horizontal center of the sign to the vertex of the angle of the curve in the right arrow shaft and to the end of the curve.
- P is the width of each arrow shaft.
- Q is the width of each arrowhead.
- R is the width of each branch in the dividing line.
- S is the corner radius.

W4-3L

- A is the length of each side of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the tip of each arrowhead and to the base of the right arrow.
- E is the vertical distance from the horizontal center of the sign to the center of the base of the left arrow.
- F is the vertical distance from the center of the base of the left arrow to the center of the base of the parallel dividing line.
- G is the vertical distance from the base of the right arrow to the base of the right dividing line.
- H is the distance from the vertical center of the sign to the tip of each arrowhead.
- J is the length of each dashed segment of the dividing line and the distance from the horizontal center of the sign to the end of the first segment.
- K is the space between the segments of the dividing line.

- L is the offset distance from the centerline of the curved shaft to the centerline of the left branch of the dividing line.
- M is the radius of the curve in the left arrow shaft
- N is the distance from the horizontal center of the sign to the vertex of the angle of the curve in the left arrow shaft and to the end of the curve.
- P is the width of each arrow shaft.
- Q is the width of each arrowhead.
- R is the width of each branch in the dividing line.
- S is the corner radius.

W4-5: Entering Roadway Merge

W4-5R

- A is the length of each side of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the vertical center of the sign to the tip of the arrowhead.
- E is the distance from the horizontal center of the sign to the tip of the arrowhead.
- F is the distance from the horizontal center of the sign to the vertex of the curve angle for the second curved section of the branch.
- G is the distance from the vertical center of the sign to the vertex of the curve angle for second curved section of the branch.
- H is radius of the second curved section of the branch.
- J is the radius of the first curved section of the branch.
- K is the shaft width of the curved branch.
- L is the distance from the vertical center of the sign to the centerline of the arrow shaft at its base.
- M is the shaft width of the straight arrow.
- N is the distance from the horizontal center of the sign to the centerline of the arrow shaft at its base.
- P is the distance from the horizontal center of the sign to the bottom of the symbol.
- Q is the width of the arrowhead.
- R is the corner radius.

W4-5L

- A is the length of each side of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the vertical center of the sign to the tip of the arrowhead.
- E is the distance from the horizontal center of the sign to the tip of the arrowhead.
- F is the distance from the horizontal center of the sign to the vertex of the curve angle for the second curved section of the branch.
- G is the distance from the vertical center of the sign to the vertex of the curve angle for second curved section of the branch.
- H is radius of the second curved section of the branch.
- J is the radius of the first curved section of the branch.
- K is the shaft width of the curved branch.
- L is the distance from the vertical center of the sign to the centerline of the arrow shaft at its base.
- M is the shaft width of the straight arrow.
- N is the distance from the horizontal center of the sign to the centerline of the arrow shaft at its base.
- P is the distance from the horizontal center of the sign to the bottom of the symbol.
- Q is the width of the arrowhead.
- R is the corner radius.

W4-6: Entering Roadway Added Lane

- A is the length of each side of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the vertical distance from the tip of the left arrowhead to the tip of the right arrowhead.
- E is the distance from the horizontal center of the sign to the tip of the right arrowhead.
- F is the distance from the horizontal center of the sign to the base of the right arrow shaft.
- G is the vertical distance from the center of the base of the left dividing line to the base of the right dividing line.
- H is the vertical distance from the center of the base of the left arrow to the center of the base of the left dividing line.
- J is the distance from the horizontal center of the sign to the center of the base of the left arrow shaft.
- K is the horizontal distance from the vertical center of the sign to the tip of the left arrowhead.
- L is the horizontal distance from the tip of the left arrowhead to the center of the end of the dividing line.
- M is the length of each dashed segment of the dividing line.
- N is the space between the dashed segments of the dividing line.
- P is the distance from the vertical center of the sign to the centerline of the vertical section of the right arrow shaft.
- Q is the radius of the curve in the centerline of the right arrow shaft.
- R is the distance from the horizontal center of the sign to the start of the curve in the right arrow shaft and to the vertex of its curve angle.
- S is the width of each arrow shaft.
- T is the horizontal distance from the center of the base of the left arrow shaft to the center of the base of the left branch of the dividing line.
- W is the width of each branch of the dividing line.
- X is the width of each arrowhead.
- Y is the corner radius.

W6-1: Divided Highway

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the tip of the right arrowhead.
- E is the distance from the horizontal center of the sign to the vertex of the first curve angle and to the start of the first curve in the right arrow shaft.
- F is the length of the first straight section of each arrow.
- G is the distance from the horizontal center of the sign to the tip of the left arrowhead.
- H is the distance from the horizontal center of the sign to the base of the left arrow shaft.
- J is the distance from the base of the left arrow shaft to the top of the traffic island symbol and to the vertex of the major curve angle for the curved edges of the traffic island.
- K is the distance from the vertical center of the sign to the left and right edges of the traffic island symbol.
- L is the radius of the major curve that forms the left- and right-side edges of the traffic island.
- M is the radius of the minor curve that forms the left- and right-bottom edges of the traffic island.
- N is the distance from the horizontal center of the sign to the vertex of the second curve angle and to the end of the second curve.
- P is the distance from the vertical center of the sign to the vertex of the second curve angle and to the end of the second curve.
- Q is the radius of the centerline of each curve in each arrow shaft.
- R is the distance from the vertical center of the sign to the tip of the left arrowhead and to the centerline of the first straight section of the right arrow shaft.
- S is the width of each arrowhead.
- T is the length of the second straight section of each arrow.
- U is the corner radius.

W6-2: Divided Highway Ends

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the tip of the left arrowhead.
- E is the distance from the horizontal center of the sign to the vertex of the first curve angle and to the start of the first curve in the left arrow shaft.
- F is the length of the first straight section of each arrow.
- G is the distance from the horizontal center of the sign to the tip of the right arrowhead.
- H is the distance from the horizontal center of the sign to the base of the right arrow shaft.
- J is the distance from the base of the right arrow shaft to the bottom of the traffic island symbol and to the vertex of the major curve angle for the side edges of the traffic island.
- K is the distance from the vertical center of the sign to the right and left edges of the traffic island symbol.
- L is the radius of the major curve that forms the right- and left-side edges of the traffic island.
- M is the radius of the minor curve that forms the right- and left-top edges of the traffic island.
- N is the distance from the horizontal center of the sign to the vertex of the second curve angle and to the end of the second curve.
- P is the distance from the vertical center of the sign to the vertex of the second curve angle and to the end of the second curve.
- Q is the radius of the centerline of each curve in each arrow shaft.
- R is the distance from the vertical center of the sign to the tip of the right arrowhead and to the centerline of the first straight section of the left arrow shaft.
- S is the width of each arrowhead.
- T is the length of the second straight section of each arrow.
- U is the corner radius.

W6-3: Two-Way Traffic

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the tip of each arrowhead.
- E is the distance from the horizontal center of the sign to the base of each arrow.
- F is the distance from the vertical center of the sign to the centerline of each arrow.
- G is the width of each arrowhead.
- H is the corner radius.

W7-1a: Hill with Grade

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the distance from the horizontal center of the sign to the top of the symbol.
- E is the distance from the horizontal center of the sign to the bottom of the symbol.
- F is the space between the symbol and the word.
- G is the letter height and FHWA standard font for the number.
- H is the distance from the vertical center of the sign to the left edge of the symbol.
- J is the distance from the vertical center of the sign to the left and right edges of the percentage.
- K is the width of the number in the percentage.
- L is the space between the number and the percent symbol in the percentage.
- M is the width of the percent symbol.
- N is the corner radius.

W7-3P: XX% Grade (plaque)

- A is the horizontal dimension of the plaque.
- B is the vertical dimension of the plaque.
- C is the inset from the edge of the plaque to the border.
- D is the border width.
- E is the distance from the top of the first line to the top of the plaque and from the bottom of the last line to the bottom of the plaque.
- F is the letter height and FHWA standard font for each line.
- G is the space between the lines.
- H is the distance from the vertical center of the plaque to the left and right edges of the percentage on the first line.
- J is the width of the number in the percentage on the first line.
- K is the space between the number and the percent symbol on the first line.
- L is the width of the percent symbol on the first line.
- M is the distance from the vertical center of the plaque to the left edge of the word on the second line.
- M is the distance from the vertical center of the plaque to the right edge of the word on the second line.
- P is the corner radius.

W7-3bP: XX% Grade XX Miles (plaque)

- A is the horizontal dimension of the plaque.
- B is the vertical dimension of the plaque.
- C is the inset from the edge of the plaque to the border.
- D is the border width.
- E is the distance from the top of the first line to the top of the plaque, the space between the lines, and the distance from the bottom of the last line to the bottom of the plaque.
- F is the letter height and FHWA standard font for each line.
- G is the distance from the vertical center of the sign to the left and right edges of the first line.
- H is the width of the number on the first line.
- J is the space between the number and the percent symbol on the first line.
- K is the width of the percent symbol on the first line.
- L is the space between the percent symbol and the word on the first line and between the number and the word on the second line.
- M is the distance from the vertical center of the plaque to the left edge of the word on the first line.
- N is the distance from the vertical center of the plaque to the right edge of the word on the first line.
- P is the distance from the vertical center of the plaque to the left and right edges of the second line.
- Q width of the number on the second line.
- R is the distance from the vertical center of the plaque to the left edge of the word on the second line.
- S is the distance from the vertical center of the plaque to the right edge of the word on the second line.
- T is the corner radius.

W12-2: Low Clearance Advance

- A is the length of each edge of the sign.
- B is the inset from the edge of the sign to the border.
- C is the border width.
- D is the height of each arrow symbol.
- E is the space between the line and each arrow symbol.
- F is the distance from the horizontal center of the sign to the top and bottom of the line.
- G is the letter height and FHWA standard font for the line.
- H is the distance from the vertical center of the sign to the left and right edges of the line.
- J is the width of the first number on the line.
- K is the space between the first number and the feet symbol and between the second number and the inches symbol.
- L is the width of the feet symbol on the line.
- M is the space between the feet symbol and the hyphen on the line.
- N is the width of the hyphen on the line.
- P is the space between the hyphen and the second number on the line.
- Q is the width of the second number on the line.
- R is the width of the inches symbol on the line.
- S is the corner radius.

W16-8P: Advance Street Name (1-line plaque)

- A is the horizontal dimension of the plaque.
- B is the vertical dimension of the plaque.
- C is the inset from the edge of the plaque to the border.
- D is the border width.
- E is the distance from the top of the line to the top of the plaque and from the bottom of the line to the bottom of the plaque.
- F is the letter height and FHWA standard font for the line.
- G is the distance from the left edge of the first word to the left edge of the plaque and from the right edge of the second word to the right edge of the plaque.
- H is the width of the first word.
- J is the space between the words.
- K is the width of the second word.
- L is the corner radius.

W16-8aP: Advance Street Name (2-line plaque)

- A is the horizontal dimension of the plaque.
- B is the vertical dimension of the plaque.
- C is the inset from the edge of the plaque to the border.
- D is the border width.
- E is the distance from the top of the first line to the top of the plaque and from the bottom of the last line to the bottom of the plaque.
- F is the letter height and FHWA standard font for the first line.
- G is the space between the words on the first line and the arrow symbol on the second line and between the words on the second line and the arrow symbol on the first line.
- H is the height of each arrow symbol.
- J is the distance from the tip of the arrowhead on the first line to the left edge of the plaque.
- K is the width of each arrow symbol.
- L is the space between the arrow and the words on each line.
- M is the width of the first word on the first line.
- N is the space between the words on each line.
- P is the width of the second word on the first line.
- Q is the distance from the right edge of the second word on the first line to the right edge of the plaque.
- R is the distance from the left edge of the first word on the second line to the left edge of the plaque and from the right edge of the first word on the second line to the right edge of the plaque.
- S is the width of the first word on the second line.
- T is the width of the second word on the second line.
- U is the corner radius.